



Description of Features

1. Introduction

Hopscotch 4-All (H4-All) is an engaging, equitable, accessible, and open-access adaptive educational recommender system that teaches research design to high school students and undergraduate college students. H4-All uses AI technologies to deliver personalized, adaptive instruction to strengthen students' research literacy and college readiness.

H4-All augments instruction by recommending next steps, scaffolds, and supports that help students successfully engage in research design. Built on the award-winning Hopscotch platform, used by more than 200,000 learners worldwide, and on Hopscotch 4-AP Research (40,000 users), H4-All combines proven inquiry-based pedagogy with large language models (LLMs) to strengthen academic mastery, critical thinking, and learner agency.

The core purpose of H4-All is to expand opportunity for all students to receive the transformative benefits of learning research design, such as critical thinking skills, by making research instruction more responsive, transparent, and supportive. Currently, participation and success in research courses remain inequitable, with students from minoritized racial and ethnic backgrounds significantly underrepresented.

Traditional research instruction often assumes prior academic capital and provides limited individualized support, reinforcing existing disparities. While emerging AI technologies offer promise for personalization at scale, they can exacerbate inequity without intentional, equity-centered design. H4-All directly addresses this challenge by embedding inclusive pedagogy into its AI-driven recommendations, expanding student success to include habits of inquiry, confidence, and self-direction within a supportive learning environment.

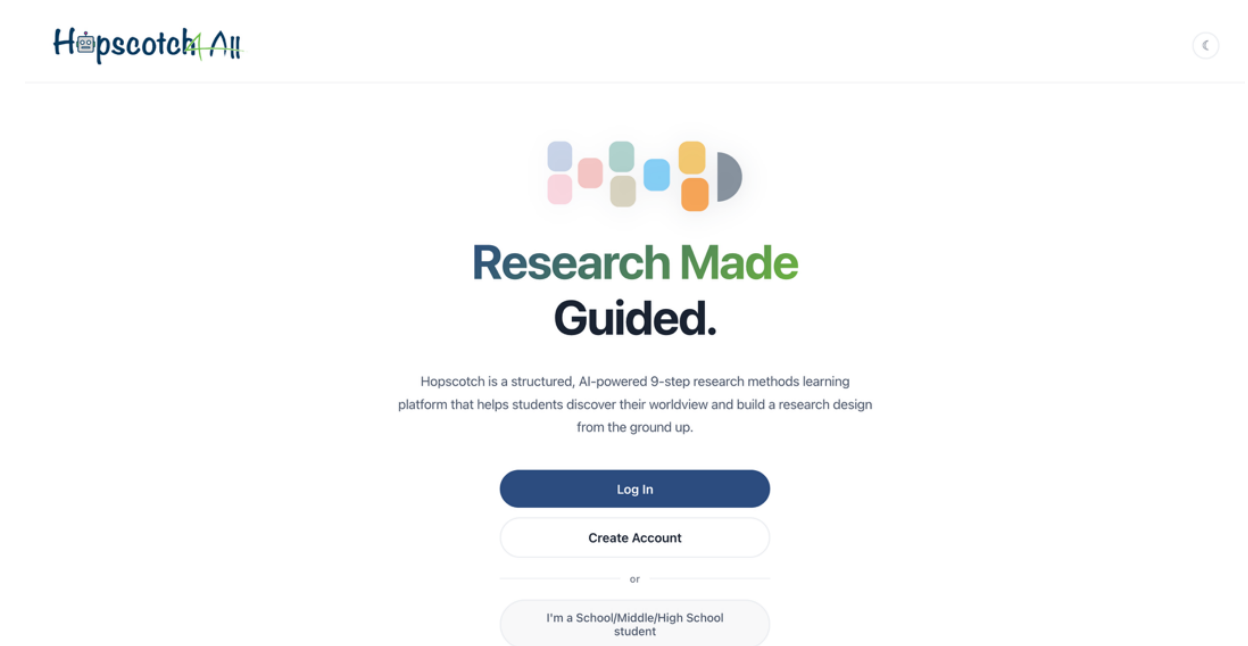
This document describes the main features of Hopscotch 4-All.

2. Creation of an Account

H4-All incorporates a multi-role account architecture consisting of:

- High School Teachers
- High School Students
- Higher Education Faculty
- Higher Education Students
- Super-Users (Administrative Role)

Teachers and faculty can create classes of unlimited size and generate student accounts using generic, non-identifiable credentials to protect the privacy of minors. The system supports structured classroom management with secure access controls.



[← Back](#)

Create Account

Sign up to get started with Hopscotch

I am a...

High School Student

High School Teacher

Higher Ed Student

Higher Ed Faculty

Full Name

Full name

Email ID

Email ID

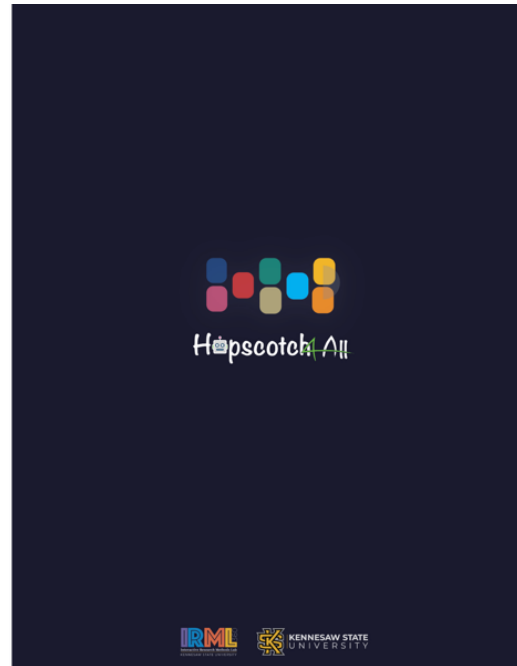
Password

Password



Create Account

Already have an account? [Login](#)



3. Features for High School Teachers

Teachers and faculty are provided with a comprehensive dashboard that enables:

- Real-time monitoring of student progress.
- Review and individualized feedback on student research designs.
- Download of student research designs in PDF format.
- Class creation and management functionality.
- Access to the same nine-step structured research design generator used by students.
- AI-assisted research design generation for demonstration or modeling purposes.

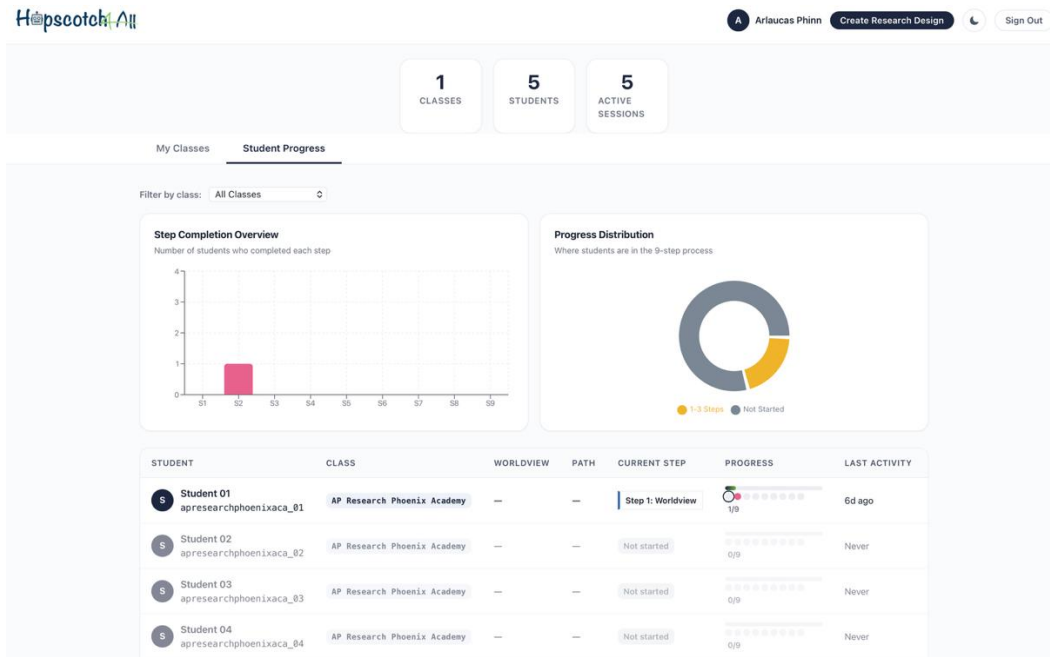
Easy creation of classes with as many students as needed. Teachers create accounts of their students (minors) with generic, unidentifiable credentials.

The screenshot shows the Hopscoth dashboard with the user 'Arlaucas Phinn' logged in. At the top, there are three summary cards: 1 CLASS, 5 STUDENTS, and 0 ACTIVE SESSIONS. Below these, the 'My Classes' tab is selected. The main content area features a 'Create a New Class' form with the following fields:

- CLASS NAME:** A text input field with the placeholder 'e.g. Period 3 Research'.
- STUDENTS:** A numeric input field set to '10'.
- SHARED PASSWORD:** A text input field with the placeholder 'Password for all students'.

A 'Create Class' button is located below the form. Below the form, the 'Your Classes' section shows a single class: 'AP Research Phoenix Academy' with 5 students, last updated on 2/19/2026.

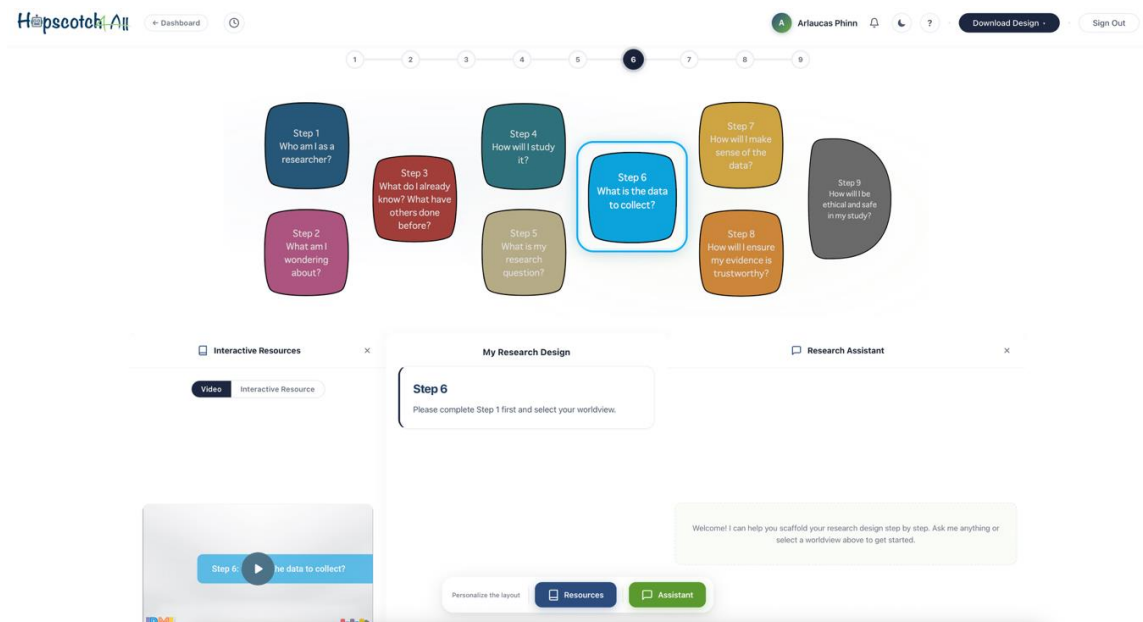
Teachers have access to a dashboard to monitor the progress of their students.



Teachers can provide individualized feedback to the research designs generated by their students. The research designs of all students can be downloaded as a pdf document.

The screenshot shows the 'Worldview' step of a research design process. On the left, a vertical sidebar lists nine steps: 1. Worldview (selected), 2. Topic & Goals, 3. Literature, 4. Methodology, 5. Research Question, 6. Data Collection, 7. Analysis, 8. Trustworthiness, and 9. Ethics. The main area is titled 'Step 1: Worldview' and contains a text input field with the text 'WORLDVIEW Constructivist'. Below this, a 'Feedback' section shows 'No feedback yet.' and a text box with the instruction 'Please make sure that the worldview that you have selected aligns well with the topic and methods of your study.' A 'Send Feedback' button is located at the bottom right of the feedback section. At the top right, there are buttons for 'Back', 'Download PDF', and 'Sign Out'.

Teachers can also generate research designs (middle panel) by following the nine steps proposed by Hopscotch (top panel), using the videos and interactive resources offered for each step (left panel), and using AI assistance for the Research Assistant (left panel)



4. Features for High School Students

High school students engage with a structured workspace organized around nine sequential research design steps.

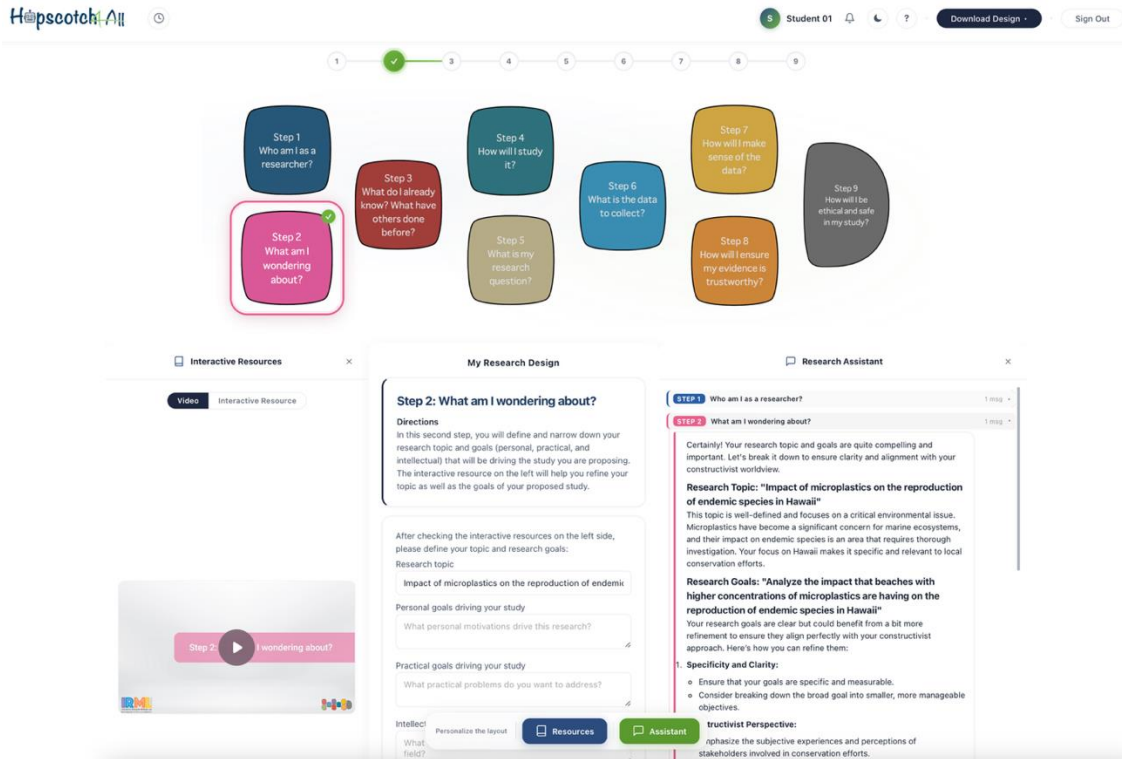
The interface consists of:

- Top Panel: Displays the nine-step research framework progression.
- Middle Panel: Active research design drafting workspace.
- Left Panel: Instructional resources including videos and interactive resources.
- Right Panel: AI Research Assistant.

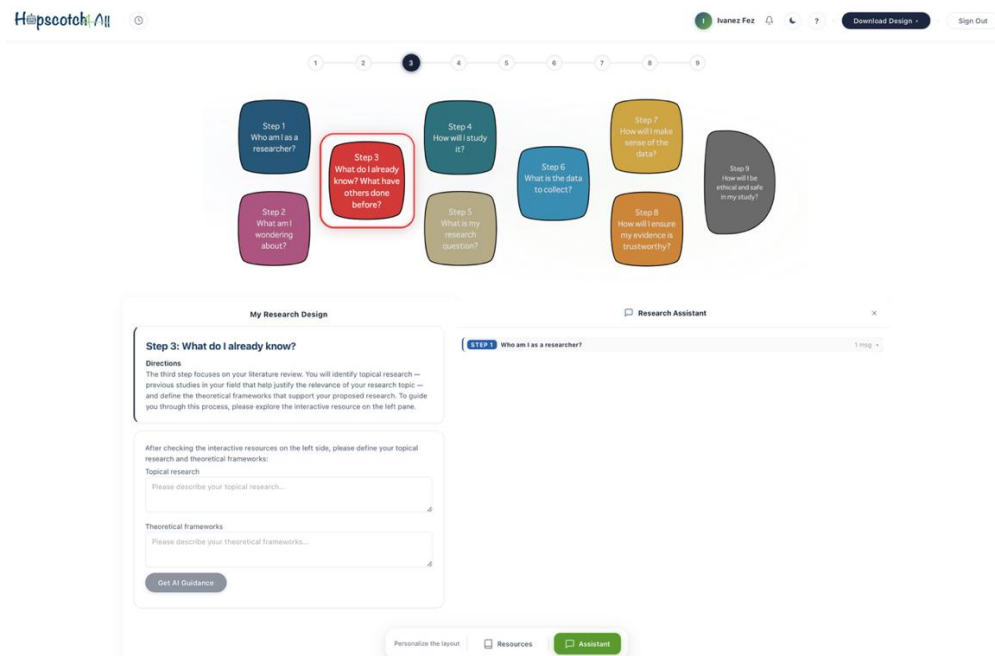
Key Capabilities:

- Generation of unlimited research designs.
- AI-assisted research guidance via contextual prompts and feedback.
- Downloadable research design outputs in PDF format.
- Auto-generated conceptual framework visualizations (three visual formats).
- Editable visualization prior to export.
- Layout personalization of workspace.
- Continuous access to instructor feedback.

Students can generate research designs (middle panel) for their projects by following the nine steps proposed by Hopscotch (top panel), using the videos and interactive resources offered for each step (left panel), and leveraging the AI assistance offered by the Research Assistant (left panel).



Students are able to personalize the layout of the working space by clicking on “personalize the layout” (lower bottom side of the interface)



Students can create as many designs as needed.

The screenshot displays the Hopscotch AI interface for creating a research design. On the left, a 'Session History' panel lists two designs: 'Untitled Research' (Feb 27, 2026, 0/9, CURRENT) and 'Use of Social Media in High ...' (Feb 27, 2026, 2/9, Qualitative). The main workspace features a flowchart with nine steps: Step 1 (Who am I as a researcher?), Step 2 (What am I wondering about?), Step 3 (What do I already know? What have others done before?), Step 4 (How will I study it?), Step 5 (What is my research question?), Step 6 (What is the data to collect?), Step 7 (How will I make sense of the data?), Step 8 (How will I ensure my evidence is trustworthy?), and Step 9 (How will I be ethical and safe in my study?). Below the flowchart, a 'My Research Design' section provides directions for Step 1, including a paragraph about paradigmatic positioning and a selection of worldviews. A 'Research Assistant' panel on the right offers a welcome message and a 'Download Design' button. The interface also includes a 'Personalize the layout' button and a 'Resources' button.

Students can download their generated research design as a pdf, as well as a visual representation (three visual options) of the conceptual framework emerging out of their designs.

This screenshot shows the Hopscotch AI interface with a completed research design. The flowchart steps are the same as in the previous image, but Step 3 is highlighted with a red border and a green checkmark. The 'Download Design' button is now active, and a dropdown menu shows two options: 'PDF Research Design' and 'Conceptual Framework'. The 'Research Assistant' panel on the right is still visible, and the 'Personalize the layout' button is present at the bottom.

Before downloading the visual representation of the conceptual framework emerging from their designs, students can edit it at their pleasure.

Students can check at any given time the feedback provided by their teachers.

5. Features for Higher Education Faculty

Easy creation of classes with as many students as needed. Faculty can create accounts of their students with generic, unidentifiable credentials.

H@pscotch AI Ivan Abellan [Create Research Design](#) [Sign Out](#)

1 CLASSES

10 STUDENTS

0 ACTIVE SESSIONS

[My Classes](#) [Student Progress](#)

Create a New Class

Student accounts will be auto-generated with the shared password you set below.

CLASS NAME
e.g. Period 3 Research

STUDENTS
10

SHARED PASSWORD
Password for all students

Create Class

Your Classes 1

EDRS9100

edrs9100

10 students

2/18/2026

CLASS CODE

PASSWORD

STUDENTS

edrs9100

EDRS9100-SUMMER26

10

edrs9100_01

edrs9100_02

edrs9100_03

edrs9100_04

edrs9100_05

edrs9100_06

edrs9100_07

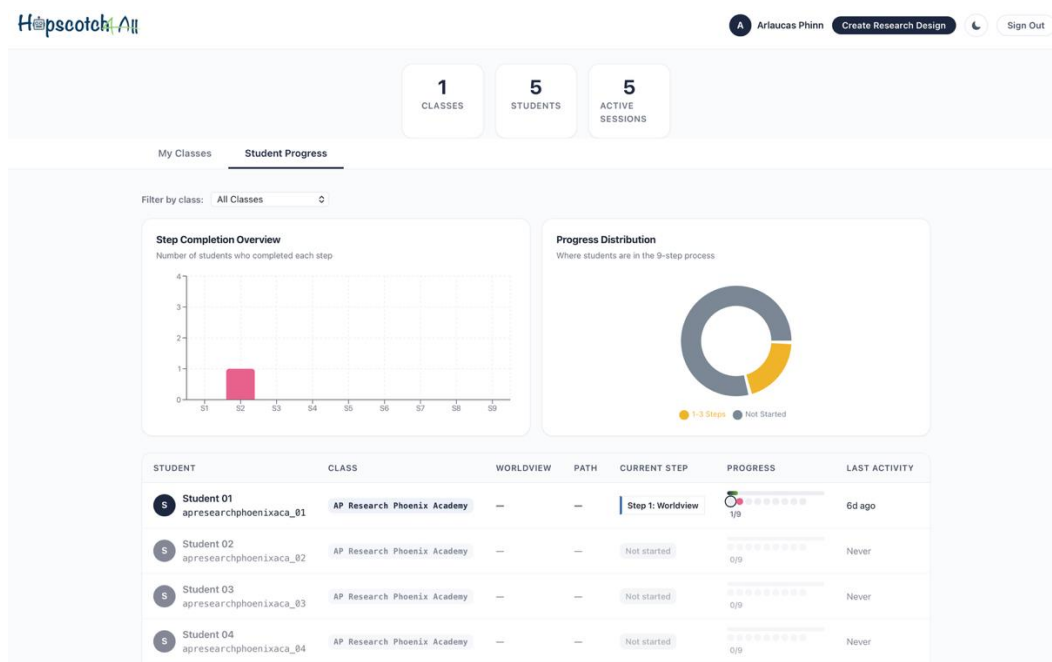
edrs9100_08

edrs9100_09

edrs9100_10

Print Credentials

Faculty have access to a dashboard to monitor the progress of their students.



Faculty can provide individualized feedback to the research designs generated by their students. The research designs of all students can be downloaded as a pdf document.

[← Back](#) [apresearchwalton_01](#) [AP-Research-Walton](#) [Constructivist](#) [Qualitative](#) [Download PDF](#)

✓ Worldview

✓ Topic & Goals

3 Literature

4 Methodology

5 Research Question

6 Data Collection

7 Analysis

8 Trustworthiness

9 Ethics

Step 1: Worldview

WORLDVIEW
Constructivist

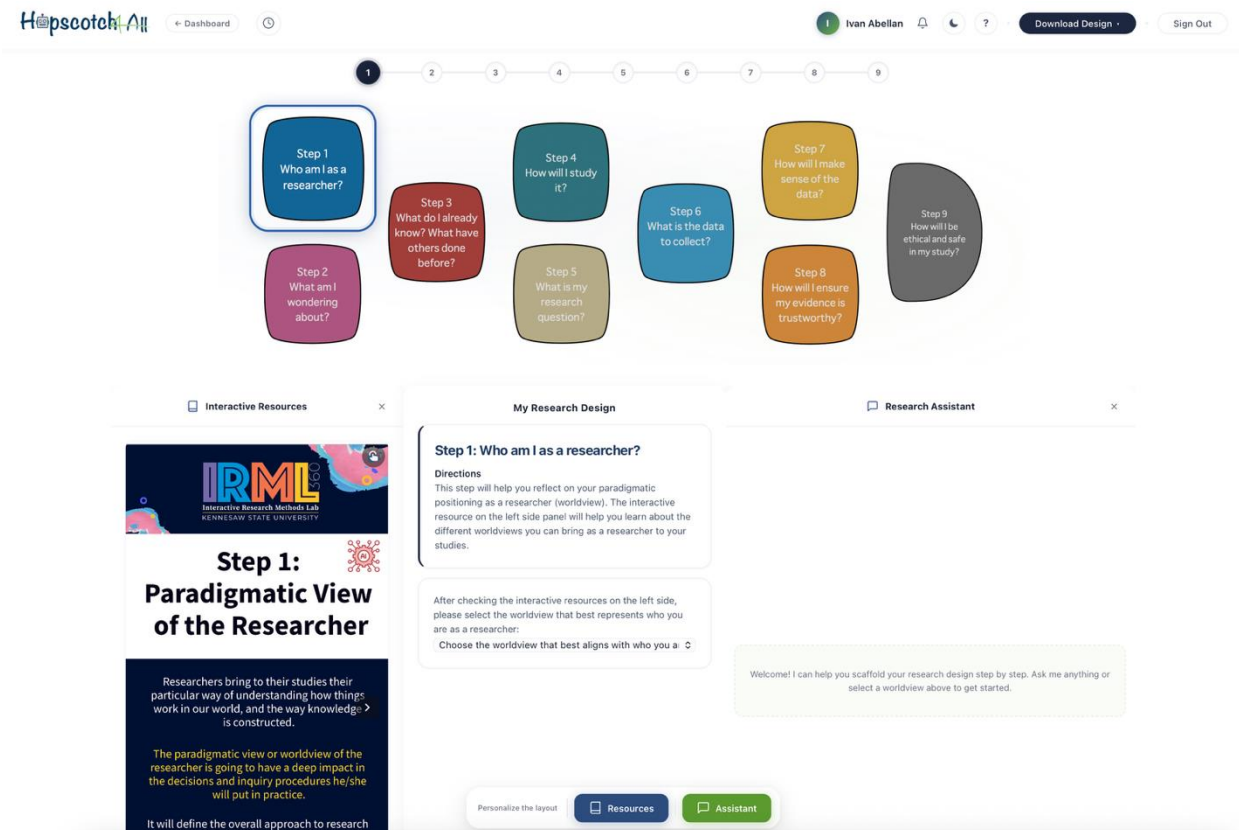
Feedback

No feedback yet.

Please make sure that the worldview that you have selected aligns well with the topic and methods of your study.

Send Feedback

Faculty can also generate research designs (middle panel) by following the nine steps proposed by Hopscotch (top panel), using the interactive resources offered for each step (left panel), and using AI assistance for the Research Assistant (left panel).



6. Features for Higher Education Students

Higher Education students engage with a structured workspace organized around nine sequential research design steps.

The interface consists of:

- Top Panel: Displays the nine-step research framework progression.
- Middle Panel: Active research design drafting workspace.
- Left Panel: Instructional resources including interactive modules.
- Right Panel: AI Research Assistant.

Key Capabilities:

- Generation of unlimited research designs.
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- Downloadable research design outputs in PDF format.
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- Editable visualization prior to export.
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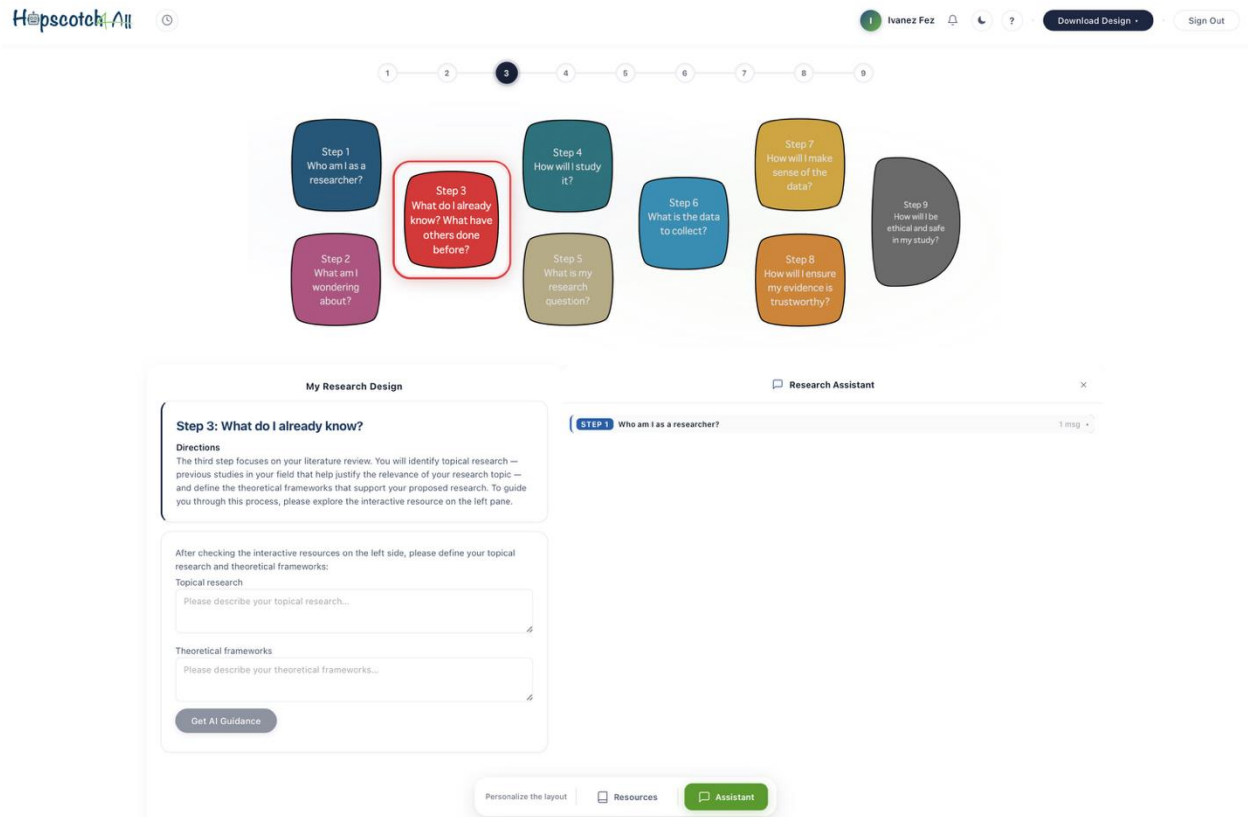
Students can generate research designs (middle panel) for their capstone projects by following the nine steps proposed by Hopscotch (top panel), using the interactive resources offered for each step (left panel), and leveraging the AI assistance offered by the Research Assistant (left panel).

The screenshot displays the Hopscotch AI interface, which is designed to guide students through a nine-step research design process. At the top, a progress bar shows steps 1 through 9, with step 2 currently selected and highlighted in green. Below the progress bar, a central area contains nine colored boxes, each representing a step: Step 1 (Who am I as a researcher?), Step 2 (What am I wondering about?), Step 3 (What do I already know? What have others done before?), Step 4 (How will I study it?), Step 5 (What is my research question?), Step 6 (What is the data to collect?), Step 7 (How will I make sense of the data?), Step 8 (How will I ensure my evidence is trustworthy?), and Step 9 (How will I be ethical and safe in my study?).

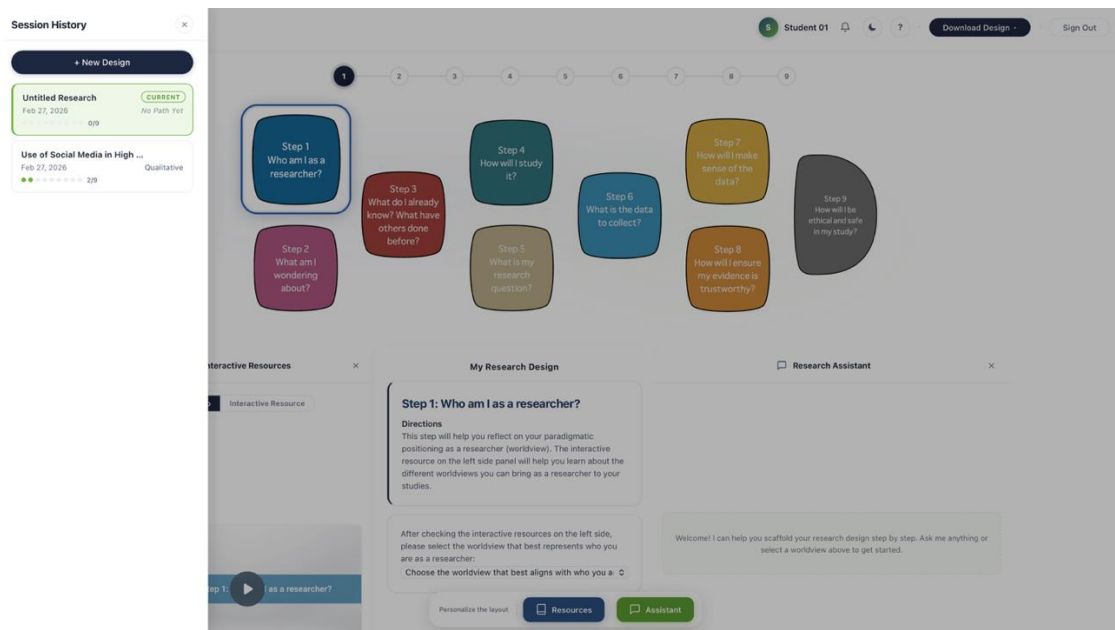
Below the central area, there are three main panels:

- Interactive Resources:** This panel on the left provides resources for each step. It includes a video player showing a video titled "Step 2: What am I wondering about?" and a list of interactive resources.
- My Research Design:** This central panel displays the current step's instructions and goals. For Step 2, it includes "Directions" (defining and narrowing down the research topic and goals), "Research topic" (Impact of microplastics on the reproduction of endemic), "Personal goals driving your study" (What personal motivations drive this research?), "Practical goals driving your study" (What practical problems do you want to address?), and "Intellect" (Personalize the layout).
- Research Assistant:** This panel on the right provides AI assistance. It includes a chat interface where the user can ask questions. The assistant's response for Step 2 includes a "Research Topic: 'Impact of microplastics on the reproduction of endemic species in Hawaii'" and "Research Goals: 'Analyze the impact that beaches with higher concentrations of microplastics are having on the reproduction of endemic species in Hawaii'". It also provides a "Constructivist Perspective" and a list of "Specificity and Clarity" guidelines.

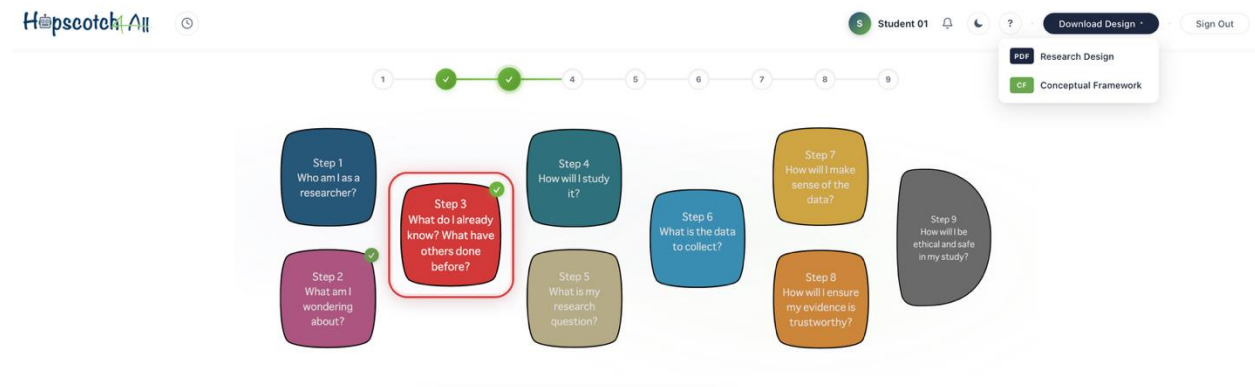
Students are able to personalize the layout of the working space by clicking on “personalize the layout” (lower bottom side of the interface)



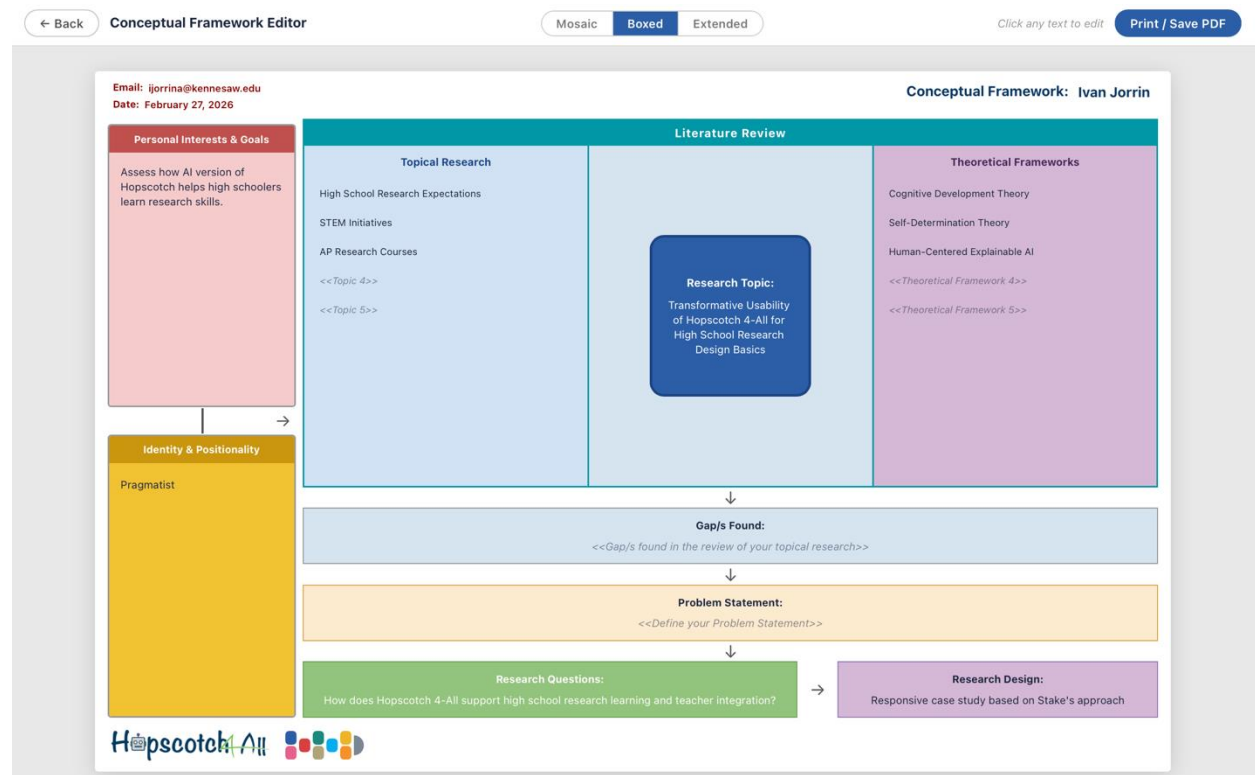
Students can create as many designs as needed.



Students can download their generated research design as a pdf, as well as a visual representation (three visual options) of the conceptual framework emerging out of their designs.



Before downloading the visual representation of the conceptual framework emerging out of their designs, students can edit it at their pleasure.



Students can check at any given time the feedback provided by their teachers.

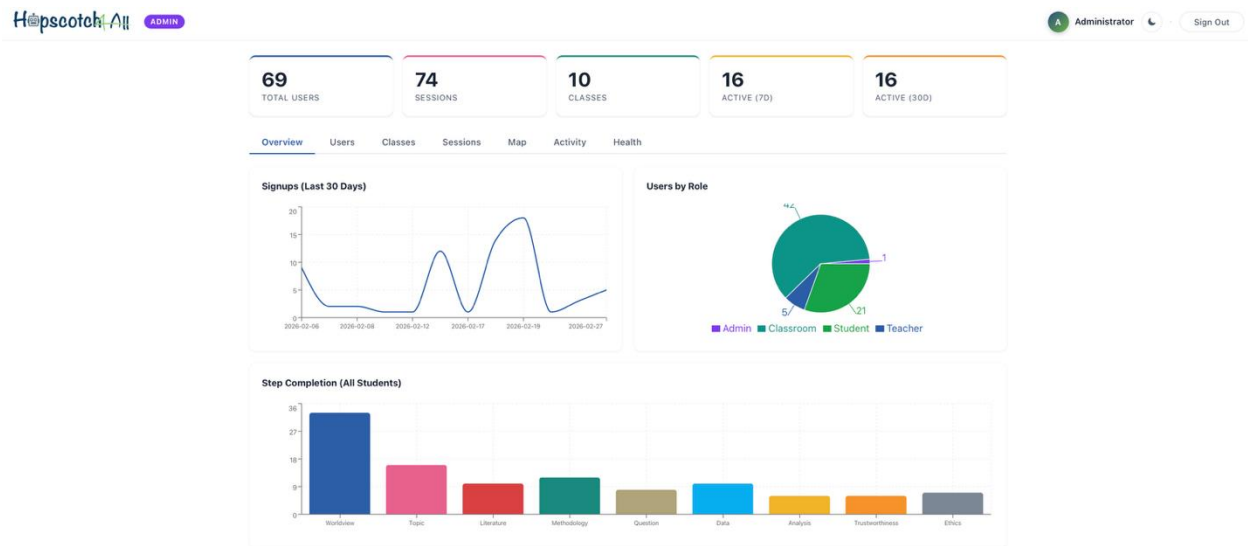
The screenshot displays the Hopscotch AI interface for a student named 'Student 01'. At the top, a progress bar shows steps 1 through 9, with steps 1 and 2 marked as complete. The main area features a grid of nine steps, each in a colored box with a question: Step 1 (blue, 'Who am I as a researcher?'), Step 2 (pink, 'What am I wondering about?'), Step 3 (red, 'What do I already know? What have others done before?'), Step 4 (teal, 'How will I study it?'), Step 5 (olive, 'What is my research question?'), Step 6 (blue, 'What is the data to collect?'), Step 7 (yellow, 'How will I make sense of the data?'), Step 8 (orange, 'How will I ensure my evidence is trustworthy?'), and Step 9 (grey, 'How will I be ethical and safe in my study?'). Step 1 is highlighted with a blue border and a green checkmark. Below the steps, there are three panels: 'Interactive Resources' on the left with a video player for 'Step 1: Who am I as a researcher?'; 'My Research Design' in the center, which includes a 'Directions' section, a paragraph about paradigmatic positioning, a dropdown menu for 'Constructivist', and buttons for 'Personalize the layout', 'Resources', and 'Assistant'; and 'Research Assistant' on the right, which provides detailed text about the research process. A 'Teacher Feedback' panel on the top right shows a message from 'Arlaucas Phinn' dated 'Just now'.

7. Features for Super-users

Super-users have access to platform-wide analytics and administrative controls, including:

- Monitoring total users.
- Monitoring total sessions.
- Tracking number of classes.
- Active user tracking (7-day and 30-day metrics).
- User management.
- Class management.
- Session management.
- Geolocation analytics for sessions and users.

Analytics Dashboard



Super-users can also manage users, classes, sessions, and geolocate the sessions and users.

Hopscootch AI ADMIN Administrator Sign Out

69
TOTAL USERS

74
SESSIONS

10
CLASSES

16
ACTIVE (7D)

16
ACTIVE (30D)

Overview **Users** Classes Sessions Map Activity Health

Search by name, email, or username... All Roles Export CSV Create User

USER	ROLE	EDUCATION	STATUS	LAST LOGIN	IP	CREATED	ACTIONS
Student 05 apresearchwalton_05	CLASSROOM	High School	Active	—	—	2/27/2026	🔍 🗑️ 📄
Student 04 apresearchwalton_04	CLASSROOM	High School	Active	—	—	2/27/2026	🔍 🗑️ 📄
Student 03 apresearchwalton_03	CLASSROOM	High School	Active	—	—	2/27/2026	🔍 🗑️ 📄
Student 02 apresearchwalton_02	CLASSROOM	High School	Active	—	—	2/27/2026	🔍 🗑️ 📄
Student 01 apresearchwalton_01	CLASSROOM	High School	Active	30m ago	138.218.12.38	2/27/2026	🔍 🗑️ 📄
Student 03 cs2803_03	CLASSROOM	High School	Active	—	—	2/26/2026	🔍 🗑️ 📄

69

TOTAL USERS

74

SESSIONS

10

CLASSES

16

ACTIVE (7D)

16

ACTIVE (30D)

Overview Users **Classes** Sessions Map Activity Health

Search by class name or code...

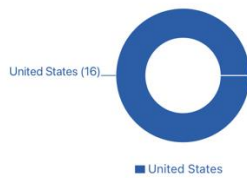
CLASS NAME	CODE	TEACHER	STUDENTS	PASSWORD	CREATED	ACTIONS
AP-Research-Walton	apresearchwalton	Arlaucas Phinn highschoolteacher@kennesaw.edu	5 / 5	apwaltonspring26	2/27/2026	
CS2003	cs2003	Mourya Mourya@test.com	3 / 3	cs2003students	2/26/2026	
classresearch	classresearch	Mourya Mourya@test.com	6 / 6	classresearch	2/19/2026	
AP Research Phoenix Academy	apresearchphoenixaca	Arlaucas Phinn highschoolteacher@kennesaw.edu	5 / 5	apresearchphoenix	2/19/2026	
periodresearch	periodresearch	Mourya Mourya@test.com	2 / 2	periodresearch123	2/19/2026	
EDRS9100	edrs9100	Ivan Abellan ivanjerrin140@gmail.com	10 / 10	EDRS9100-SUMMER26	2/18/2026	
test1	test1	Mourya Mourya@test.com	4 / 4	-	2/13/2026	
test	test	Mourya Mourya@test.com	3 / 3	-	2/13/2026	
resa	resa	Mourya Mourya@test.com	2 / 2	-	2/13/2026	
elementary1	elementary1	Mourya Mourya@test.com	2 / 2	-	2/13/2026	

Previous Page 1 of 1 (10 classes) Next

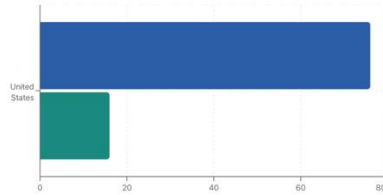
User Login Locations



Users by Country



Logins by Country



Top Cities / Regions

